

MINI PROJECT 1

AWS Web Server Deployment

with VPC, EC2, Nginx and Security Best Practices

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Course: Cloud Computing

1. Introduction

This project involved the end-to-end design, deployment, and documentation of a secure, internet-facing web server on Amazon Web Services (AWS). The objective was to apply practical cloud engineering skills including VPC networking, EC2 provisioning, Linux administration, Nginx configuration, and AWS Identity and Access Management (IAM).

The server was deployed inside a custom Virtual Private Cloud (VPC) with a public subnet, internet gateway, and properly configured route tables. A Security Group enforcing the principle of least privilege controls all inbound traffic, and SSH access is restricted to the engineer's IP address only.

For the bonus task, the default Nginx page was replaced with a personal DevOps portfolio page showcasing the AWS skills demonstrated throughout this project.

2. Architecture

2.1 Architecture Diagram

The following describes the logical flow of traffic from the public internet to the Nginx web server:

Internet Users → **Internet Gateway** → **Route Table** → **Public Subnet** →
Security Group → **EC2 (Nginx)**

(See attached architecture-diagram.png for the visual diagram)

2.2 AWS Resources Summary

Resource	Value / Configuration	Purpose
VPC	10.0.0.0/16	Isolated network environment
Public Subnet	10.0.1.0/24 (us-east-1a)	Hosts public-facing resources
Internet Gateway	Attached to VPC	Enables internet connectivity
Route Table	0.0.0.0/0 → IGW	Routes internet traffic through IGW
Security Group	Ports 22, 80, 443	Controls inbound/outbound traffic
EC2 Instance	t2.micro, Ubuntu 22.04	Virtual server running Nginx
IAM User	EC2FullAccess + VPCReadOnly	Least-privilege admin access

3. Deployment Steps

3.1 Network Infrastructure Setup

- Navigated to VPC → Your VPCs → Create VPC
- Set Name: project-vpc, IPv4 CIDR: 10.0.0.0/16
- Created Public Subnet: Name: public-subnet-1, CIDR: 10.0.1.0/24, AZ: us-east-1a
- Created Internet Gateway (project-igw) and attached it to project-vpc
- Created Route Table (public-rt), added route 0.0.0.0/0 → project-igw
- Associated public-subnet-1 with public-rt
- Enabled auto-assign public IP on the subnet

3.2 Security Group Configuration

- Created Security Group: web-server-sg attached to project-vpc
- Added inbound rule: Port 22 (SSH) — Source: My IP (xxx.xxx.xxx.xxx/32)
- Added inbound rule: Port 80 (HTTP) — Source: 0.0.0.0/0
- Added inbound rule: Port 443 (HTTPS) — Source: 0.0.0.0/0
- Outbound: All traffic allowed (default)

3.3 EC2 Instance Launch

- AMI: Ubuntu Server 22.04 LTS (HVM), SSD Volume Type
- Instance type: t2.micro (Free Tier eligible)
- Network: project-vpc | Subnet: public-subnet-1
- Auto-assign public IP: Enabled
- Security Group: web-server-sg
- Key pair: Created new key pair (project-key.pem), downloaded securely
- Launched instance and confirmed status: Running

3.4 SSH Connection

Connected to the instance from a local terminal:

```
chmod 400 project-key.pem
ssh -i project-key.pem ubuntu@<PUBLIC_IP>
```

3.5 Nginx Installation and Configuration

- `sudo apt update && sudo apt upgrade -y`
- `sudo apt install nginx -y`
- `sudo systemctl start nginx`
- `sudo systemctl enable nginx`
- Verified service: `sudo systemctl status nginx`

4. Security Measures

4.1 Security Group Rules

Port	Protocol	Source	Justification
22	TCP	My IP only	Prevents brute-force SSH attacks from the public internet
80	TCP	0.0.0.0/0	Required for public HTTP web access
443	TCP	0.0.0.0/0	Required for HTTPS (TLS) web access

4.2 IAM User Configuration

- Created IAM user: cloud-project-admin
- Attached policy: AmazonEC2FullAccess (for EC2 management)
- Attached policy: AmazonVPCReadOnlyAccess (for VPC inspection)
- Enabled MFA on the IAM user account
- Root account MFA also enabled; root access not used for day-to-day tasks

4.3 SSH Key Security

- Key pair file (project-key.pem) stored securely locally
- `chmod 400` applied to prevent unauthorized reads
- SSH access restricted to a single source IP in the Security Group

5. Troubleshooting Log

The following issues were encountered during deployment, along with their causes and resolutions.

Issue 1: SSH Connection Timed Out

Problem	The ssh command hung indefinitely with no response and eventually timed out.
Cause	The route table had not been associated with the public subnet. Without this association, the subnet had no route to the internet even though the Internet Gateway existed.
Solution	Navigated to VPC → Route Tables → selected public-rt → Subnet Associations tab → clicked Edit subnet associations → selected public-subnet-1 → saved. SSH connected successfully on the next attempt.

Issue 2: Permission Denied (publickey)

Problem	Running the ssh command returned: Permission denied (publickey). The connection was refused immediately without prompting for any credentials.
Cause	The downloaded .pem key file had permissions set to 644 (readable by all users). OpenSSH enforces strict permission checks and rejects key files that are accessible by other users as a security measure.
Solution	Ran chmod 400 project-key.pem in the terminal to restrict the file to owner-read-only. Re-ran the ssh command and connected successfully.

Issue 3: Web Page Not Loading in Browser

Problem	After installing Nginx and confirming it was running with systemctl status, opening the public IP in a browser showed This site can't be reached.
Cause	The Security Group attached to the EC2 instance only had an inbound rule for port 22 (SSH). Port 80 (HTTP) had not been added, so all browser traffic was being blocked at the Security Group level before reaching Nginx.
Solution	Opened EC2 → Security Groups → selected web-server-sg → Inbound rules → Edit inbound rules → added a new rule: Type = HTTP, Port = 80, Source = 0.0.0.0/0. Refreshed the browser and the Nginx page loaded immediately.

6. Lessons Learned

- VPC networking is the foundation of AWS security — every resource needs explicit routes, subnets, and gateway associations to communicate.
- Security Groups are stateful: allowing inbound traffic automatically permits the return traffic, so only inbound rules need to be carefully crafted.

- The principle of least privilege is not just theoretical — restricting SSH to My IP only and using a non-root IAM user meaningfully reduces the attack surface.
- Route table associations are easy to miss but critical: without associating the route table to the subnet, internet traffic cannot flow even if the IGW exists.
- Linux systemctl commands (start, enable, status) are essential for reliable service management — enable ensures the server restarts on reboot automatically.
- chmod 400 is required before any SSH attempt with .pem files; OpenSSH enforces strict key permission checks as a security measure.
- Deploying a custom portfolio page reinforced how Nginx serves static files from /var/www/html and how file ownership and permissions affect web serving.

7. Bonus Task — Personal DevOps Portfolio Page

7.1 Overview

For the bonus task, the default Nginx welcome page was replaced with a custom personal portfolio page. The page is designed to function as a DevOps profile, showcasing the AWS and Linux skills demonstrated throughout this project. It is served directly by Nginx from the EC2 instance and is publicly accessible via the instance's public IP address.

7.2 Steps Taken

- SSH'd into the EC2 instance using the project key pair
- Navigated to the Nginx web root: `cd /var/www/html`
- Opened the default index file for editing: `sudo nano index.html`
- Replaced all default Nginx content with a custom HTML/CSS portfolio page
- Set correct file ownership: `sudo chown www-data:www-data index.html`
- Set correct permissions: `sudo chmod 644 index.html`
- Reloaded Nginx to apply changes: `sudo systemctl reload nginx`
- Verified the portfolio page loaded correctly in the browser via the public IP

7.3 Portfolio Page Content

The custom portfolio page includes the following sections:

- Header with name, title (Junior Cloud Engineer), and contact links
- About Me section describing cloud engineering background and goals
- Skills section listing AWS services covered in this project: VPC, EC2, IAM, Security Groups, Nginx
- Projects section highlighting this AWS deployment as a featured project
- Deployment metadata: public IP address and deployment date

7.4 Key Commands

```
cd /var/www/html  
sudo nano index.html  
sudo chown www-data:www-data index.html  
sudo chmod 644 index.html  
sudo systemctl reload nginx
```

7.5 Result

The portfolio page is live and accessible at http://<PUBLIC_IP>. It replaces the default Nginx page entirely and demonstrates both Linux file management and web content deployment skills. A screenshot of the page rendered in the browser is included in the project screenshots folder.

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